Cheatsheet AZ-104 — 03 · Governance: subscriptions, management groups, Policy, locks, tags, costs and Advisor

1. Hierarchy: tenant, management groups, subscriptions and resource groups

Level	What it is	Exam facts
Tenant (Entra ID)	Directory that the subscriptions trust	A subscription trusts **a single** tenant. Changing its tenant = transfer and **redo RBAC** (assignments do not travel)
Tenant Root Group	Root management group; ID = tenant ID; cannot be moved or deleted	Nobody has access by default: a Global Admin **elevates**. Anything assigned here (RBAC or Policy) affects **everything**
Management groups	Containers of subscriptions and other MGs to apply RBAC and Policy by inheritance	**10,000** per tenant, **6 levels** (not counting root or subscriptions), **a single parent**. New subscriptions land in the root (or in the configured "default management group"). "Require authorization" prevents anyone from creating MGs
Subscription	Unit of billing, quotas and trust boundary	**980** resource groups. Types: Free, Pay-As-You-Go, EA, MCA, CSP, **Dev/Test** (reduced rates, no Windows / SQL license charges, non-production only). Cancel → resources disabled, reactivable for **90 days**. You must **register resource providers** (`Microsoft.Compute...`) before using a new type
Resource group	Logical container; its region only stores metadata; resources from several regions	A resource lives in **one** RG; deleting the RG deletes everything; **800** resources per type. Moving resources: same tenant, source and destination RGs **locked for up to 4 h**, the resource's RBAC assignments **are lost**, validate first, a ReadOnly lock on the source or destination RG blocks the move. Changing subscription: VM + disks + NIC + VNet **together**

- **Moving a subscription between MGs:** write on the source and destination MG + `roleAssignments/write` and `delete` on the subscription (that is, subscription **Owner**; if you inherit it from the source MG, you lose it when moving). If the source or destination is the root, no permissions on it are needed.
- **Billing:** EA = Enrollment → Departments → Accounts → Subscriptions (roles: Enterprise Administrator, Department Administrator, Account Owner). MCA = Billing account → Billing profiles → Invoice sections → Subscriptions. These roles are **not** RBAC: a Billing account owner is not Owner of the resources.
- **Subscription policies** (portal → Subscriptions → Manage Policies; only a Global Administrator with elevate access; since May 2026 both default to "Allow no users"): prevent subscriptions from **leaving** the tenant or **entering** from another; list of exempted users.
- **Quotas** per region and VM family (vCPU): "QuotaExceeded" → Usage + quotas → **Request increase** (do not switch subscriptions).

2. Azure Policy

Piece	What it is	Detail
Definition	JSON with `if` (condition) and `then` (effect); parameters; mode **All** or **Indexed** (only resources with tags/location)	Built-in or custom; created in an MG or subscription (definition scope)
Initiative (policy set)	Group of definitions assigned as a single one	E.g.: Microsoft Cloud Security Benchmark, ISO 27001, "Enable Azure Monitor for VMs"
Assignment	Definition or initiative + **scope** (MG, subscription, RG) + **exclusions** + parameter values + enforcement (Enabled / **DoNotEnforce** for testing)	If the effect is Modify or DeployIfNotExists it needs a **managed identity** with permissions (Contributor) at the scope
Exemption	Excludes a resource or scope from an assignment with reason **Waiver** or **Mitigated** and an expiration date	Different from exclusion: the exemption is done after assigning and is audited
Remediation task	Applies DeployIfNotExists / Modify to **already existing** resources	Policy **does not fix** existing resources on its own: it only flags them non-compliant
Compliance	Evaluation: when a resource is **created or updated**, on **assignment** (≈ 30 min), every **24 h**, or on demand (`Start-AzPolicyComplianceScan`, `az policy state trigger-scan`)	States: Compliant, Non-compliant, Exempt, Conflicting, Not started

Effect	What it does	Built-in example
Deny	Blocks create or update; error `RequestDisallowedByPolicy`	Allowed locations · Allowed resource types · Not allowed resource types · Allowed virtual machine size SKUs · Require a tag and its value
Audit	Allows, but flags non-compliant	Audit VMs that do not use managed disks
AuditIfNotExists	Audits if a **related** resource is missing (extension, diagnostic setting)	Audit VMs without the antimalware extension
DeployIfNotExists	Deploys the missing related resource (MI + remediation)	Deploy diagnostic settings to Log Analytics · Deploy Azure Monitor Agent
Modify	Adds, replaces or removes **tags** or properties on create/update and in remediation	**Inherit a tag from the resource group** · Add a tag to resources
Append	Adds fields to the request (e.g. a network rule)	Append IP rules to storage accounts
DenyAction	Blocks an action, today **DELETE** (complement to locks)	Do not allow deletion of resource types
Manual	Compliance attested by hand	Regulatory compliance controls
Disabled	To switch off a definition without removing the assignment	—

- **Policy ≠ RBAC:** Policy decides **what can be done with resources** (properties, locations, SKUs) and applies **even to an Owner**; RBAC decides **who** can do what. "Users must not create VMs larger than D4" is not solved with RBAC.
- Order: disabled → append / modify → deny → audit → manual → auditIfNotExists → denyAction; AINE / DINE are evaluated after the resource is created (≈ 10 min). Several assignments accumulate: the **most restrictive** wins.

- **Allowed locations** uses mode Indexed and usually excludes global resources; **assigning at an MG** applies to all subscriptions under it. Policy for **tags** = Modify (add / inherit) or Deny (require). Role: **Resource Policy Contributor**.

3. Resource locks

Lock	Portal	Effect
CanNotDelete	Delete	Read and modify yes; **delete no**
ReadOnly	Read-only	Read-only for **all roles**, including Owner (equivalent to Reader)

- Inherited (subscription → RG → resource); the **most restrictive** wins. Cannot be placed on management groups. They affect the **control plane**: a lock on the storage account does **not** prevent deleting blobs.
- Who manages them: **Owner** and **User Access Administrator** (`Microsoft.Authorization/locks/*`). **Contributor cannot**.
- Surprise effects: ReadOnly on a storage account → cannot **list keys** or create containers from the portal · ReadOnly on an RG with VMs → **they cannot be started or restarted** (POST) · ReadOnly on an RG with an App Service plan → cannot **scale** · ReadOnly on an RG → cannot **move** resources in or out · ReadOnly on the subscription → Advisor stops working · CanNotDelete on the Azure Backup RG (`AzureBackupRG_*`) → **backups fail** (it cannot clean up recovery points) · CanNotDelete prevents deleting RBAC assignments · a locked resource makes the deletion of its **entire** RG fail.
- Commands: `New-AzResourceLock -LockLevel CanNotDelete -LockName NoDelete -ResourceGroupName RG1` (or `-ResourceName vm1 -ResourceType Microsoft.Compute/virtualMachines`), `Get-AzResourceLock`, `Remove-AzResourceLock -LockId <id>` · `az lock create --name NoDelete --lock-type ReadOnly --resource-group RG1 [--resource-name app1 --resource-type Microsoft.Web/sites]`, `az lock list`, `az lock delete --name NoDelete --resource-group RG1`.

4. Tags

Rule	Value
Maximum	**50** pairs per resource, RG or subscription; name **512** characters (**128** on storage accounts), value **256**
Case	Name **insensitive**, value **sensitive**. Not allowed in the name: `< > % & \ ? /`
Where	Resources, RGs and subscriptions; **not** on management groups or classic resources. Some types only allow 15 (Automation, CDN, DNS zones, saved searches)
Inheritance	**Not inherited** from the RG or the subscription → Policy "Inherit a tag from the resource group" (Modify) or "Require a tag" (Deny)
Permissions	**Tag Contributor** (`Microsoft.Resources/tags/write`, even without access to the resource) or Contributor on the resource. Not Reader
Billing	Tags column in the usage file; Cost analysis **group by tag**; they only count from the moment they exist (**not retroactive**) and in services that support them

- PowerShell: `Update-AzTag -ResourceId <id> -Tag @{Owner="Ana"} -Operation Merge` (Merge adds, Replace replaces, Delete removes) · `New-AzTag -ResourceId <id> -Tag @{Env="Prod"}` (replaces everything) · `Set-AzResourceGroup -Name RG1 -Tag @{Env="Prod"}` (replaces) · `(Get-AzResource -Name vm1).Tags`.
- CLI: `az tag update --resource-id <id> --operation Merge --tags Owner=Ana` · `az tag create --resource-id <id> --tags Env=Prod` · `az group update -n RG1 --set tags.Env=Prod` · `az resource tag --tags Env=Prod --ids <id>` (replaces unless `--is-incremental`) · `az resource list --tag Env=Prod`.

5. Costs: Cost Management + Billing and Advisor

Tool	What it does	Exam details
Cost analysis	Explore actual or amortized costs; **group by** resource, RG, service, location, **tag**; filters; saved views; forecast	Needs Reader or **Cost Management Reader** at the scope. Data lags **8-24 h**; a new subscription takes up to 48 h
Budgets	Amount per period (monthly, quarterly, annual; "billing month" on PAYG) with alerts	Up to **5 thresholds** (% of the budget) and 5 emails; alerts on **actual** or **forecasted** cost; **action group** only at subscription or RG; evaluated every **24 h**; **does not stop** spending or resources; Owner creates, Contributor / Cost Management Contributor create their own, Reader only views
Cost alerts	Budget, **credit** (EA / MCA), department quota (EA) and **anomaly** alerts	Shown in Cost Management → Cost alerts
Exports	Scheduled dump (daily, weekly, monthly) to a storage account	"analyze in Power BI", "retain cost data"
Pricing calculator / **TCO calculator**	Estimate the cost of an architecture / compare it with on-prem	"estimate before deploying" / "compare with datacenter"
Savings	**Reservations** (1 or 3 years, up to ~72 %) · **Savings plan** (hourly compute spend, flexible) · **Spot** (up to ~90 %, evictable) · **Azure Hybrid Benefit** (Windows / SQL licenses with Software Assurance) · Dev/Test pricing · auto-shutdown · right-size	"predictable 3 years" → reservation; "existing licenses" → Hybrid Benefit; "interruptible" → Spot
Advisor	Recommendations in 5 categories: **Cost, Security, Reliability, Operational Excellence, Performance** + Advisor score	Cost: **shut down or resize underutilized VMs** (looks at **7 days**, configurable 7-90; shut down if P95 CPU < 3 % and outbound network < 2 %), unattached disks, unassociated public IPs, idle gateways, recommended reservations. Can be **Postponed / Dismissed**; **alerts** for new recommendations with an action group; weekly digest. Reader to view, Contributor to postpone. **Does not act on its own**

- "Stop spending when the budget is reached" **does not exist**: a budget only notifies; to act, action group → Automation runbook / Logic App that shuts down VMs.

- "Group costs by department" → **tags** + Cost analysis; if the tags did not exist before, earlier costs do **not** show up grouped.
- "Prevent users from creating expensive VMs" → Policy (Allowed virtual machine size SKUs), not a budget or RBAC.

6. Commands that show up

Task	PowerShell (Az)	Azure CLI
Management groups	`New-AzManagementGroup -GroupName mg-prod -DisplayName "Prod" -ParentId <id>` · `New-AzManagementGroupSubscription -GroupName mg-prod -SubscriptionId <id>`	`az account management-group create --name mg-prod --display-name Prod --parent <id>` · `az account management-group subscription add --name mg-prod --subscription <id>`
Subscriptions and providers	`Get-AzSubscription` · `Set-AzContext -Subscription "Prod"` · `Register-AzResourceProvider -ProviderNamespace Microsoft.Compute`	`az account list -o table` · `az account set -s "Prod"` · `az provider register --namespace Microsoft.Compute` · `az provider list --query "[?registrationState=='Registered']"`
Resource groups	`New-AzResourceGroup -Name RG1 -Location westeurope` · `Remove-AzResourceGroup -Name RG1 -Force` · `Get-AzResourceGroup -Tag @{Env="Prod"}`	`az group create -n RG1 -l westeurope` · `az group delete -n RG1 --yes --no-wait` · `az group list --tag Env=Prod`
Move resources	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <id> [-DestinationSubscriptionId <id>]` · validate: `Invoke-AzResourceAction -Action validateMoveResources`	`az resource move --destination-group RG2 --ids <id> [--destination-subscription-id <id>]` · `az resource invoke-action --action validateMoveResources`
Policy	`New-AzPolicyAssignment -Name loc -Scope <id> -PolicyDefinition (Get-AzPolicyDefinition -BuiltIn ...) -PolicyParameterObject @{listOfAllowedLocations=@("westeurope")}` · `Get-AzPolicyState -Filter "ComplianceState eq 'NonCompliant'"` · `Start-AzPolicyRemediation`	`az policy assignment create --name loc --scope <id> --policy <def> --params '{"listOfAllowedLocations": {"value":["westeurope"]}}'` · `az policy state list --filter "complianceState eq 'NonCompliant'"` · `az policy remediation create`
Locks and tags	See sections 3 and 4	See sections 3 and 4
Budgets	`New-AzConsumptionBudget -Amount 500 -Name b1 -Category Cost -TimeGrain Monthly -StartDate ... -NotificationKey Key1 -NotificationEnabled -NotificationThreshold 80 -ContactEmail ...`	`az consumption budget create --amount 500 --budget-name b1 --category cost --time-grain monthly --start-date ... --end-date ...`

7. Decision tree

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What does the wording ask for?
├─ Same RBAC or Policy across many subscriptions ..... Management group
├─ Prevent creating resources outside a region ..... Policy: Allowed locations (Deny)
├─ Block expensive VM SKUs ..... Policy: Allowed virtual machine size SKUs
├─ Require a tag on creation ..... Policy: Require a tag and its value (Deny)
├─ Add the RG's tag to its resources ..... Policy: Inherit a tag from the RG (Modify) + remediation
├─ Fix already existing resources ..... Remediation task (DeployIfNotExists / Modify)
├─ Prevent accidental deletions ..... Lock CanNotDelete (Owner / UAA)
├─ Freeze a resource completely ..... Lock ReadOnly (beware VM start/restart)
├─ Notify when spending exceeds X ..... Budget (+ action group)
├─ View costs by department ..... Tags + Cost analysis (group by tag)
├─ Find underutilized VMs ..... Advisor (Cost)
├─ Move resources to another RG or subscription ..... Move (same tenant; redo RBAC)
├─ Move resources to another region ..... Azure Resource Mover (not a move)
└─ "Subscription not registered for resource type" ..... Register the resource provider
```

8. Signals → answer

Wording	Answer
"apply the policy to all current and future subscriptions"	Assign at the **management group** (or at the root)
"resources can only be created in West Europe"	Allowed locations with effect Deny
"existing non-compliant resources remain", "no resources were changed"	Normal Policy behavior; to fix them, remediation
"users should be unable to delete the resource group", "even Owners"	Lock CanNotDelete
"VM cannot be started after applying a lock"	ReadOnly on the RG: switch to CanNotDelete
"cannot list storage keys in the portal"	ReadOnly lock on the account (or Reader role)
"cannot move the resource"	ReadOnly lock on source or destination; dependencies not included; different tenant; provider not registered
"role assignments missing after moving"	Recreate them: they do not move
"notify when cost exceeds 80 %", "forecasted"	Budget with actual or forecasted alert
"stop the VMs when the budget is reached"	Budget → action group → Automation runbook (the budget only notifies)
"charge back costs to departments"	Mandatory tags (Policy) + Cost analysis
"tags not applied to new resources in the resource group"	RG tags are not inherited: Policy Modify
"identify underutilized VMs", "resize recommendations"	Advisor → Cost
"the subscription must trust a different tenant"	Transfer subscription → change directory; RBAC is lost
"move subscription to another management group" fails	Missing write on source/destination MG or Owner on the subscription
"QuotaExceeded when creating a VM"	Request quota increase for that family and region

Wording	Answer
"minimize cost for a 3-year predictable VM"	Reserved instance (+ Hybrid Benefit if licenses exist)

9. Mistakes to avoid

1. Using RBAC to limit regions, SKUs or tags: that is **Azure Policy**.
2. Expecting Policy to fix or delete existing resources: it only flags them; fixing = remediation.
3. Believing a ReadOnly lock is harmless: it blocks starting VMs, scaling plans, listing keys and moving resources.
4. Thinking Contributor can create locks: Owner or User Access Administrator is needed.
5. Assuming resource group tags flow down to the resources.
6. Believing a budget stops spending.
7. Moving resources between subscriptions in different tenants, or leaving out the dependencies (disks, NIC, VNet).
8. Forgetting to register the resource provider in the destination subscription.
9. Creating more than 6 levels of management groups or a second parent.
10. Confusing billing roles (EA / MCA) with RBAC roles.